

Advanced Excel Guide

Comprehensive Reference Notes: XLOOKUP & Conditional Aggregations

This reference document covers essential modern Excel formulas used for data retrieval and targeted reporting. It covers the versatile XLOOKUP function followed by conditional aggregation methods (SUMIF, SUMIFS, COUNTIF, COUNTIFS, AVERAGEIF, AVERAGEIFS, MAXIFS, and MINIFS) to construct highly responsive, data-driven worksheets.

1. The XLOOKUP Function

Introduced to replace both VLOOKUP and HLOOKUP, XLOOKUP searches a range or an array and returns an item corresponding to the first match it finds. If no match exists, it can return a custom value you specify.

Syntax:

```
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])
```

Arguments Breakdown:

- lookup_value**: The value you want to search for (e.g., an ID, name, or code).
- lookup_array**: The single column or row range where Excel will execute the search.
- return_array**: The column or row range containing the data you want to retrieve.
- [if_not_found]**: *Optional*. The text/value to display if no match is found (avoids the ugly #N/A).
- [match_mode]**: *Optional*. 0 = Exact match (default), -1 = Exact or next smaller, 1 = Exact or next larger, 2 = Wildcard match.
- [search_mode]**: *Optional*. 1 = First-to-last (default), -1 = Last-to-first (reverse search).

Why XLOOKUP is Superior to VLOOKUP:

- Leftward Lookups**: The return column does not need to be to the right of the lookup column.
- Column Insertion Proof**: It references explicit ranges instead of hardcoded column index numbers, meaning inserting a column won't break your formulas.
- Exact Match by Default**: No more accidentally forgetting to type FALSE at the end.

Reference Data Table (Used for examples throughout)

Emp ID (Col A)	Employee Name (Col B)	Department (Col C)	Region (Col D)	Sales (Col E)
E101	Alice Vance	Sales	North	\$45,000
E102	Bob Miller	Marketing	South	\$28,000
E103	Charlie Smith	Sales	North	\$62,000
E104	Diana Prince	HR	East	\$15,000
E105	Evan Wright	Marketing	North	\$34,000

Example 1: Basic Left-to-Right Exact Match

Find the Department for Employee ID "E103".

```
=XLOOKUP("E103", A2:A6, C2:C6, "Not Found")
```

Result: Sales

Example 2: Leftward Lookup (VLOOKUP cannot do this cleanly)

Find the Employee ID for "Diana Prince".

```
=XLOOKUP("Diana Prince", B2:B6, A2:A6, "Missing")
```

Result: E104

2. Single-Criterion Conditional Aggregations (IF Series)

These functions evaluate a single criteria block before carrying out mathematical operations.

A. COUNTIF

Counts the number of cells within a range that meet a single specified criterion.

```
=COUNTIF(range, criteria)
```

Example: Count how many employees work in the "North" region.

```
=COUNTIF(D2:D6, "North")
```

Result: 3

B. SUMIF

Adds the values in a range that meet a single specified criterion.

```
=SUMIF(range, criteria, [sum_range])
```

Note: If [sum_range] is omitted, Excel sums the actual cells evaluated in the range argument.

Example: Sum total sales for the "Sales" department.

```
=SUMIF(C2:C6, "Sales", E2:E6)
```

Result: \$107,000 (\$45,000 + \$62,000)

C. AVERAGEIF

Calculates the average (arithmetic mean) of cells that match a given criterion.

```
=AVERAGEIF(range, criteria, [average_range])
```

Example: Calculate the average sales for the "Marketing" department.

```
=AVERAGEIF(C2:C6, "Marketing", E2:E6)
```

Result: \$31,000 ((\$28,000 + \$34,000) / 2)

3. Multi-Criteria Conditional Aggregations (IFS Series)

When you need to aggregate data based on **multiple overlapping rules**, use the "IFS" variants. **Crucial structural shift:** In the IFS series, the range to be mathematically processed (summed/averaged) is always placed as the **first argument**, unlike the single-criterion equivalent.

A. COUNTIFS

Counts cells based on multiple criteria across multiple distinct ranges.

```
=COUNTIFS(criteria_range1, criteria1, [criteria_range2, criteria2], ...)
```

Example: Count how many employees in the "Sales" department are from the "North" region.

```
=COUNTIFS(C2:C6, "Sales", D2:D6, "North")
```

Result: 2 (Alice and Charlie)

B. SUMIFS

Sums up data within a range based on multiple structural criteria boundaries.

```
=SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)
```

Example: Sum the sales where the region is "North" AND the sales are greater than \$40,000.

```
=SUMIFS(E2:E6, D2:D6, "North", E2:E6, ">40000")
```

Result: \$107,000 (Alice and Charlie match; Evan is skipped because his sales are \$34,000)

C. AVERAGEIFS

Returns the mathematical mean of all cells satisfying multiple given conditions.

```
=AVERAGEIFS(average_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)
```

Example: Calculate the average sales for employees in the "North" region with sales exceeding \$30,000.

```
=AVERAGEIFS(E2:E6, D2:D6, "North", E2:E6, ">30000")
```

Result: \$47,000 ((\$45,000 + \$62,000 + \$34,000) / 3)

4. Conditional Minimum and Maximum (MAXIFS / MINIFS)

Prior to these functions, finding conditional extrema required processing computationally heavy array formulas (Ctrl+Shift+Enter). These functions streamline the task perfectly.

A. MAXIFS

Returns the maximum value among cells specified by a given set of conditions or criteria.

```
=MAXIFS(max_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)
```

Example: Find the maximum sales recorded within the "North" region.

```
=MAXIFS(E2:E6, D2:D6, "North")
```

Result: \$62,000 (Charlie)

B. MINIFS

Returns the minimum value among cells specified by a given set of conditions or criteria.

```
=MINIFS(min_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)
```

Example: Find the minimum sales inside the "Marketing" department.

```
=MINIFS(E2:E6, C2:C6, "Marketing")
```

Result: \$28,000 (Bob)

5. Visual Cheat Sheet: Syntactical Layout Differences

Function Class	Order of Parameters	Logical Rule Handled
SUMIF / AVERAGEIF	(Criteria_Range, Criteria, [Math_Range])	Single condition check; target math range is placed at the end.
SUMIFS / AVERAGEIFS MAXIFS / MINIFS	(Math_Target_Range, Criteria_Range1, Criteria1, ...)	Multiple conditions; mathematical execution field must be placed first .
COUNTIF / COUNTIFS	(Criteria_Range1, Criteria1, ...)	No standalone independent math range required (only evaluations on conditions).